飞兆半导体参考设计

下文参考设计支持针对FEB-L013的LED照明设计。应当用在演示板上，与FL7930B、FAN7621S的数据手册以及飞兆半导体应用笔记配合使用，有问题请咨询飞兆半导体技术支持团队。请访问飞兆半导体网站：http://www.fairchildsemi.com。

应用	飞兆半导体器件	输入电压范围	输出功率	输出电压（额定电流）
LED SMPS	FAN7930B FAN7621S FCPF11N60NT FDPF7N60NZ	85-277VAC@300VAC$_{MAX}$	150W	103V(1.46A)

主要功能

- 具有次级侧反馈电路的恒压(CV)和恒流(CC)控制
- 以PFC零电流开关和DC-DC转换器零电压开关实现低EMI
- 准确可调的输出过压保护
- 内部软启动和无过冲控制
- 内部总谐波失真(THD)优化器
- 为半桥谐振转换器拓扑提供了50%占空比的变频控制
- 固定死区时间(350ns)
- 工作频率最高可达300kHz
- 通过外部LV$_{CC}$为所有保护提供自动重启操作
- 低栅极电荷
- 低Crss
- 快速开关
- 100%经过雪崩测试
- 改善的dv/dt能力
- 提高静电放电保护能力
- 保护功能 UVLO、过压、过流

1. 原理图

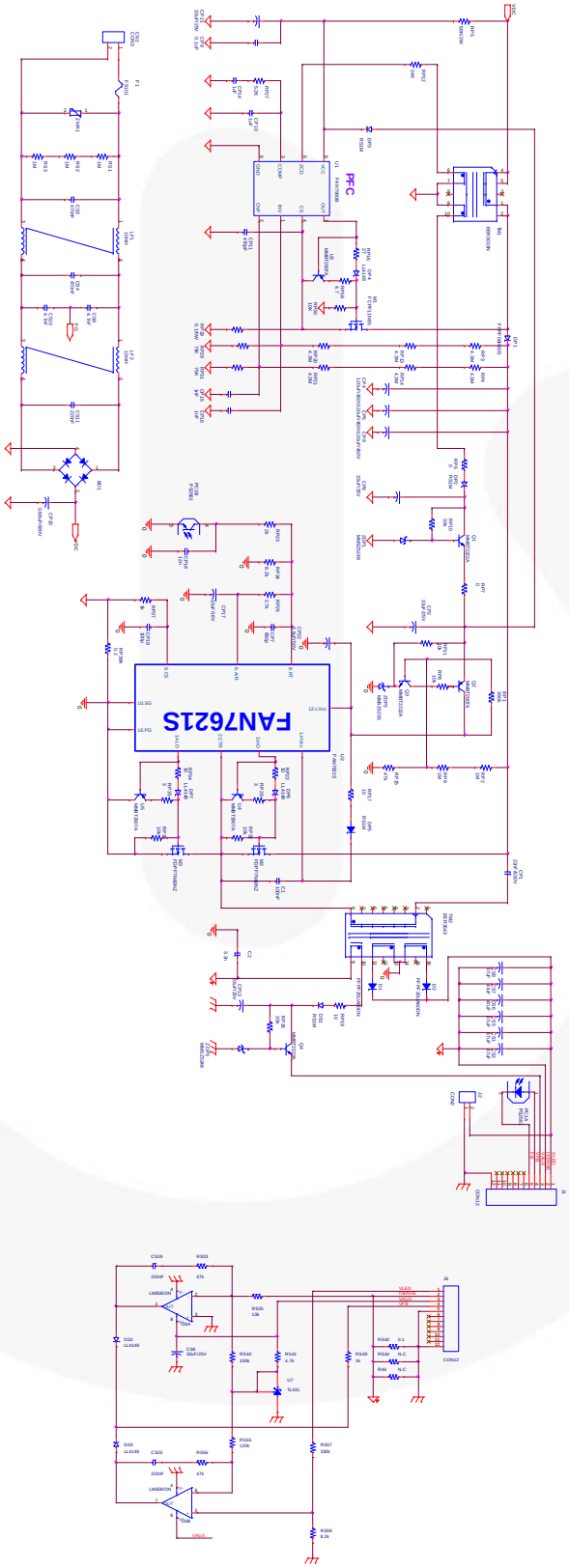

图 1. 原理图

2. 变压器

2.1 PFC变压器和绕组连接规格

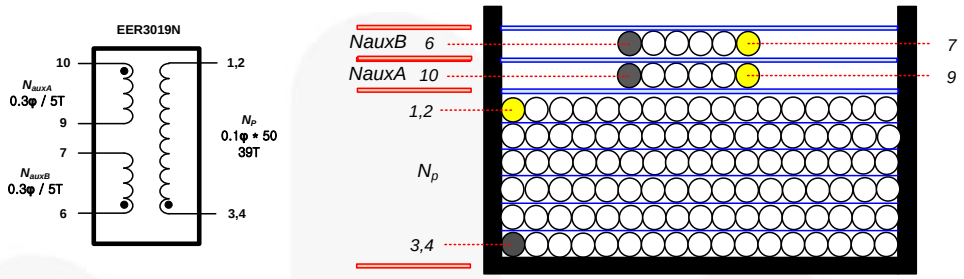

图 2. 变压器规格和结构

表 1. 绕组规格

否	绕组	引脚(S → F)	绕线	匝数	绕线方式
1	Np	3,4 → 1,2	0.1φ×50	39 Ts	螺线管绕制
2	绝缘： 聚酯带，t = 0.025mm，3 层				
3	NauxA	10 → 9	0.3φ	5 Ts	螺线管绕制
4	绝缘： 聚酯带，t = 0.025mm，3 层				
5	NauxB	6→ 7	0.3φ	5Ts	螺线管绕制
6	绝缘： 聚酯带，t = 0.025mm，3 层				

表 2. 电气特性

	引脚	规格	备注
电感量	3,4 → 1,2	194uH ±5%	100KHz， 1V

2.2 PFC变压器和绕组连接规格

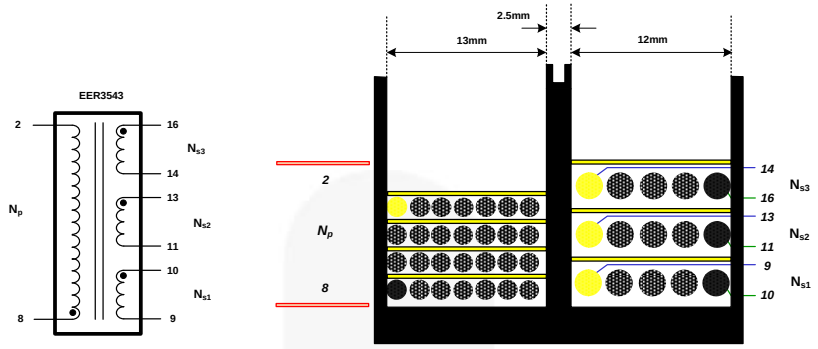

图 3. 变压器规格和结构

表 3. 绕组规格

否	绕组	引脚(S → F)	绕线	匝数	绕线方式
1	Np	8 → 2	0.1φ×20	36Ts	螺线管绕制
2	绝缘： 聚酯带，t = 0.025mm，3 层				
3	Ns1	10 → 9	0.3φ	3 Ts	螺线管绕制
4	绝缘： 聚酯带，t = 0.025mm，3 层				
5	Ns2	13→ 11	0.1φ×20	19 Ts	螺线管绕制
6	绝缘： 聚酯带，t = 0.025mm，3 层				
7	Ns3	16→ 14	0.1φ×10	19Ts	中心螺线管绕制
8	绝缘： 聚酯带，t = 0.025mm，3 层				

表 4. 电气特性

	引脚	规格	备注
初级电感（Lp）	2 – 8	690uH ±5%	100KHz， 1V
初级端有效漏电感(LR)	2 – 8	122uH（最大值）	短路次级侧的一组绕组

2.1. 绕组规格

编号	绕组	引脚(S → F)	绕线	匝数	绕线方式
1	Np1	4 → 3	0.25 Φ * 1	50	螺线管绕制
2	绝缘： 聚酯带，t = 0.05 mm，3层				
3	Ns	10 → 9	0.32 Φ (TEX) * 1	22	螺线管绕制
4	绝缘： 聚酯带，t = 0.05 mm，3层				
5	Np2	3 → 5	0.25 Φ * 1	22	螺线管绕制
6	绝缘： 聚酯带，t = 0.05 mm，3层				
7	Naux	2 → 1	0.20 Φ * 1	11	中心螺线管绕制
8	外层绝缘 聚酯带，t = 0.05 mm，3层				

2.2. 电气特性

	引脚	规格	备注
电感量	4 – 5	0.59mH ± 10%	1kHz， 1V

3. 性能

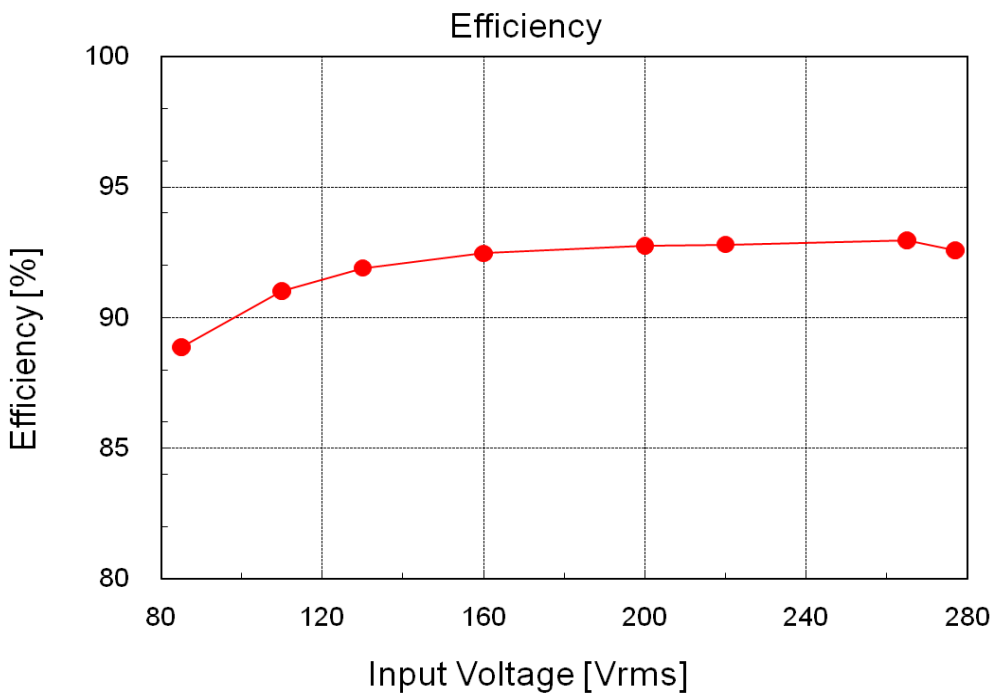

图 4. 效率曲线

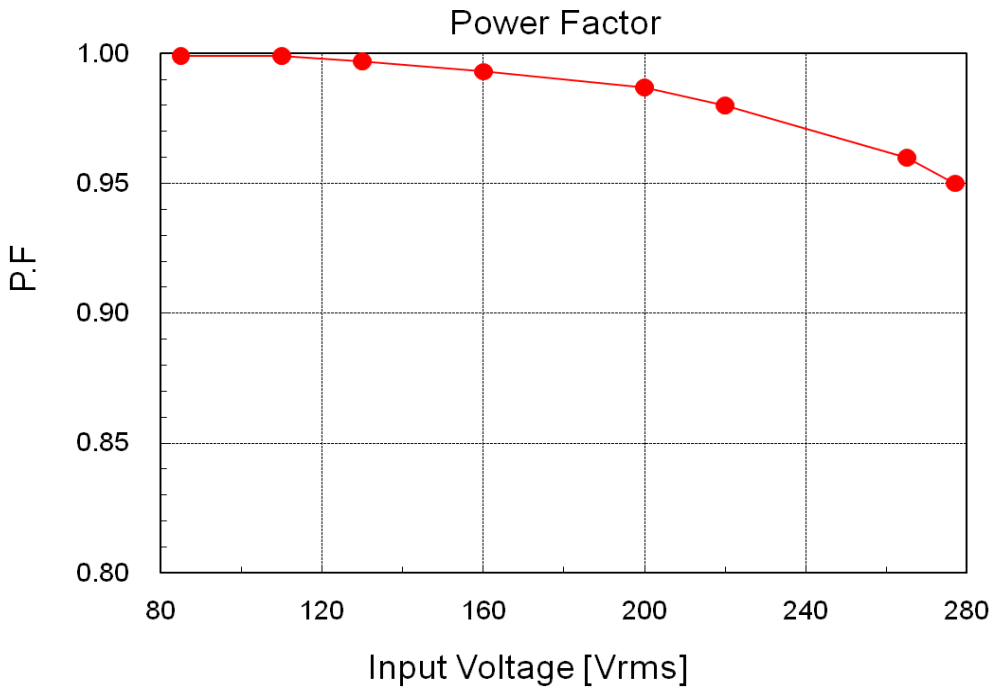

图 5. 功率因数

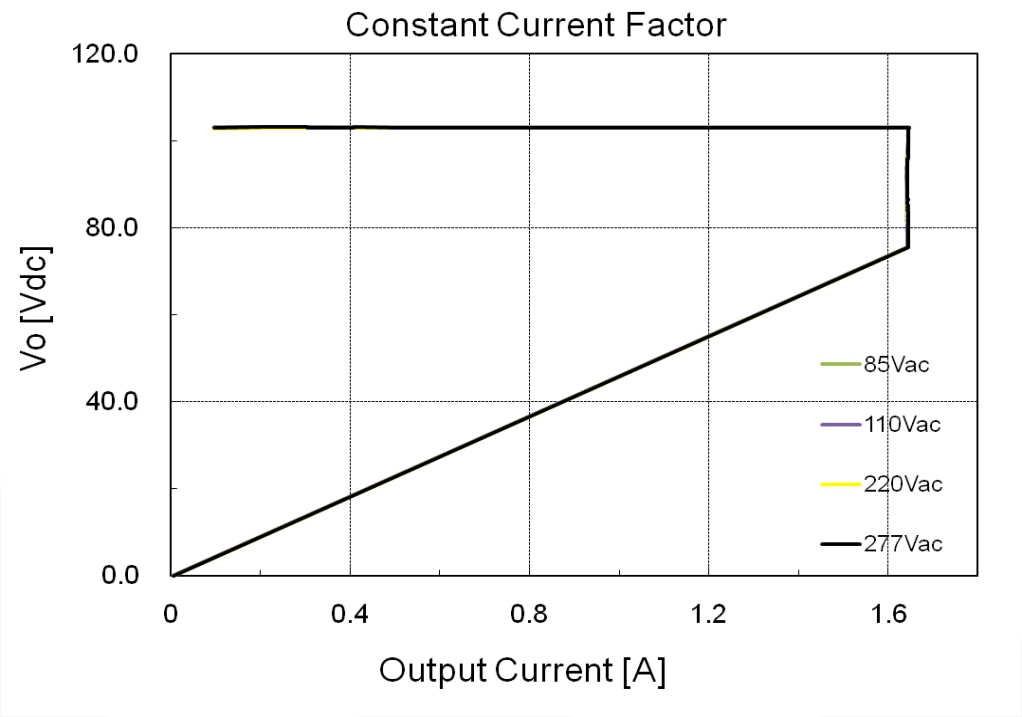

图 6. C.C/C.V曲线

4. 相关资源

数据手册链接：FAN7930B

数据手册链接：FAN7621S

数据手册链接：FCPF11N60NT

数据手册链接：FDPF7N60NZ

http://www.fairchildsemi.com/referencedesign/

Reference Design Disclaimer

Fairchild Semiconductor Corporation ("Fairchild") provides these reference design services as a benefit to our customers. Fairchild has made a good faith attempt to build for the specifications provided or needed by the customer. Fairchild provides this product "as is" and without "recourse" and MAKES NO WARRANTY, EXPRESSED, IMPLIED OR OTHERWISE, INCLUDING ANY WARRANTY OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE.

Customer agrees to do its own testing of any Fairchild reference designs in order to ensure design meets the customer needs. Neither Fairchild nor Customer shall be liable for incidental or consequential damages, including but not limited to, the cost of labor, requalifications, rework charges, delay, lost profits, or loss of goodwill arising out of the sale, installation or use of any Fairchild product.

Subject to the limitations herein, Fairchild will defend any suit or proceeding brought against Customer if it is based on a claim that any product furnished hereunder constitutes an infringement of any intellectual property rights. Fairchild must be notified promptly in writing and given full and complete authority, information and assistance (at Fairchild's expense) for defense of the suit. Fairchild will pay damages and costs therein awarded against Customer but shall not be responsible for any compromise made without its consent. In no event shall Fairchild's liability for all damages and costs (including the costs of the defense by Fairchild) exceed the contractual value of the products or services that are the subject of the lawsuit. In providing such defense, or in the event that such product is held to constitute infringement and the use of the product is enjoined, Fairchild, in its discretion, shall procure the right to continue using such product, or modify it so that it becomes noninfringing, or remove it and grant Customer a credit for the depreciated value thereof. Fairchild's indemnity does not extend to claims of infringement arising from Fairchild's compliance with Customer's design, specifications and/or instructions, or the use of any product in combination with other products or in connection with a manufacturing or other process. The foregoing remedy is exclusive and constitutes Fairchild's sole obligation for any claim of intellectual property infringement and Fairchild makes no warranty that products sold hereunder will not infringe any intellectual property rights.

All solutions, designs, schematics, drawings, boards or other information provided by Fairchild to Customer are confidential and provided for Customer's own use. Customer may not share any Fairchild materials with other semiconductor suppliers.